fn new_poisson_rdp_curve

Michael Shoemate

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Proves soundness of `fn new_poisson_rdp_curve` in `mod.rs`.

1 Hoare Triple

Precondition

`base_curve` upper-bounds the base Rényi divergence at every order > 1, and `mean` ≥ 0.

Pseudocode

```
def new_poisson_rdp_curve(base_curve, mean: float):
    def curve(alpha: float):
        if mean == 0.0:  #
            return 0.0
        if alpha <= 1.0:  #
            return float("inf")
        eps_alpha = base_curve(alpha)
        max_eps_lo = log_alpha_over_alpha_minus_one_lo(alpha)
        if eps_alpha > max_eps_lo:  #
            return float("inf")
        inv_alpha_hi = 1.0.inf_div(alpha)
        inv_alpha_lo = 1.0.neg_inf_div(alpha)
        one_minus_inv_alpha_hi = 1.0.inf_sub(inv_alpha_lo)
        one_minus_inv_alpha_minus_one_hi = one_minus_inv_alpha_hi.inf_sub(1.0)
        ln_one_minus_inv_alpha_hi = one_minus_inv_alpha_minus_one_hi.inf_ln_1p()
        alpha_minus_one_lo = alpha.neg_inf_sub(1.0)
        exponent_hi = alpha_minus_one_lo.inf_mul(ln_one_minus_inv_alpha_hi)
        power_hi = exponent_hi.inf_exp()
        delta_hi = inv_alpha_hi.inf_mul(power_hi)
        term1_hi = mean.inf_mul(delta_hi)
        log_mean_hi_ = log_mean_hi(mean)
        term2_hi = log_mean_hi_.inf_div(alpha_minus_one_lo)
        return eps_alpha.inf_add(term1_hi).inf_add(term2_hi)
    return curve
```

Postcondition

Theorem 1.1. For every $\alpha > 1$, the returned function either returns `Err(e)` due to a propagated helper error, or returns an upper bound on the implemented Poisson expression

$$\epsilon(\alpha) \;+\; \texttt{mean} \cdot \delta(\alpha) \;+\; \frac{\log(\texttt{mean})}{\alpha - 1},$$

where $\epsilon(\cdot)$ is the base Rényi divergence curve and

$$\delta(\alpha) = \alpha^{-1}\left(1 - \alpha^{-1}\right)^{\alpha-1}.$$

2 Proof

Fix $\alpha > 1$.

Lemma 2.1. If `mean = 0`, the returned value is 0, which equals the displayed expression. If $\alpha \leq 1$, the returned value is $+\infty$, which is a valid upper bound.

Proof. When `mean = 0`, both additive terms involving `mean` vanish. For $\alpha \leq 1$, $+\infty$ upper-bounds any real value. □

Lemma 2.2. If the admissibility check on line 9 passes, then the exact base Rényi divergence satisfies the Theorem 6 admissibility condition.

Proof. Let $\epsilon(\alpha)$ denote the exact base Rényi divergence curve. By precondition, `eps_alpha` $\geq \epsilon(\alpha)$. By the proof of `log_alpha_over_alpha_minus_one_lo`,

$$\texttt{max_eps_lo} \leq \log\left(\frac{\alpha}{\alpha - 1}\right).$$

Therefore `eps_alpha` $\leq$ `max_eps_lo` implies

$$\epsilon(\alpha) \leq \log\left(\frac{\alpha}{\alpha - 1}\right),$$

which is exactly the admissibility condition. □

Lemma 2.3. If the admissibility check on line 9 fails, the function returns $+\infty$, which is a valid upper bound on the displayed expression.

Proof. This is exactly the branch taken by the pseudocode, and $+\infty$ upper-bounds any real value. □

Lemma 2.4. If `mean > 0`, $\alpha > 1$, and the admissibility check passes, then the value `delta_hi` computed by the function satisfies

$$\texttt{delta_hi} \geq \delta(\alpha).$$

Proof. We have `inv_alpha_hi` $\geq 1/\alpha$ and `inv_alpha_lo` $\leq 1/\alpha$. Hence

$$\texttt{one_minus_inv_alpha_hi} \geq 1 - \frac{1}{\alpha},$$

so after subtracting 1,

$$\texttt{one_minus_inv_alpha_minus_one_hi} \geq -\frac{1}{\alpha}.$$

Both values lie in $(-1, 0)$, so upward-rounded `ln_1p` gives

$$\texttt{ln_one_minus_inv_alpha_hi} \geq \ln\left(1 - \frac{1}{\alpha}\right).$$

Also `alpha_minus_one_lo` $\leq \alpha - 1$ with `alpha_minus_one_lo` > 0. Since the logarithm above is negative, multiplying first by the smaller positive factor and then by the larger negative factor is conservative:

$$\texttt{exponent_hi} \geq (\alpha - 1)\ln\left(1 - \frac{1}{\alpha}\right).$$

Upward-rounded exponentiation yields

$$\texttt{power_hi} \geq \left(1 - \frac{1}{\alpha}\right)^{\alpha - 1}.$$

Multiplying by `inv_alpha_hi` $\geq 1/\alpha$ gives the claim. □

Lemma 2.5. If `mean > 0`, $\alpha > 1$, the admissibility check passes, and the function returns `Ok(out)`, then `out` upper-bounds the displayed Poisson expression.

Proof. By precondition, `eps_alpha` $\geq \epsilon(\alpha)$. By Lemma 2.4,

$$\texttt{term1_hi} \geq \texttt{mean} \cdot \delta(\alpha).$$

By the proof of `log_mean_hi`,

$$\texttt{log_mean_hi_} \geq \log(\texttt{mean}).$$

Since `alpha_minus_one_lo` $\leq \alpha - 1$ with a positive denominator,

$$\texttt{term2_hi} \geq \frac{\log(\texttt{mean})}{\alpha - 1}.$$

Finally, `out` is formed by upward-rounded addition of these three upper bounds. □

Proof of Theorem 1.1. By Lemma 2.1, Lemma 2.3, Lemma 2.2, and Lemma 2.5. □